ORLANDO HEALTH
1414 Kuhl Ave.
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DEPARTMENT GUIDELINE

Guideline Title: **Stroke**	Guideline Number: **GUID-3203-M017**

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Revision dates: 4/30/24	Developed by: Air Care Team
Approved by: Dr. Christopher Hunter, MD Air Care Team Medical Director	Approved by:
Signature:	Signature:
Department Numbers: 3203	

I. **PURPOSE:** To recognize the patient presenting with acute stroke or TIA; to provide and maintain adequate oxygenation, ventilation, and end-organ tissue perfusion; and to provide emergent care and expeditious transport to the nearest appropriate medical facility.

II. **DEFINITIONS:**
   a. CPP – Cerebral perfusion pressure
   b. ETCO2 – End-tidal carbon dioxide
   c. GCS – Glasgow Coma Scale
   d. ICP – Intracranial pressure
   e. LOC - Level of consciousness
   f. NIHSS – National Institutes of Health Stroke Scale
   g. TIA - Transient ischemic attack
   h. tPA – Tissue plasminogen activator (Alteplase), a thrombolytic agent

III. **PATHOPHYSIOLOGY:**
   a. Stroke should be considered in any patient presenting with an acute neurologic deficit (focal or global) or altered level of consciousness (LOC). No feature distinguishes ischemic from hemorrhagic stroke, although nausea, vomiting, headache, and change in LOC are more common in hemorrhagic strokes. Common symptoms of stroke include abrupt onset of hemiparesis, monoparesis, or quadriparesis; monocular or binocular visual loss; visual field deficits; diplopia; dysarthria; ataxia; vertigo; aphasia; or sudden decrease in LOC.
   b. Establishing the time of onset of symptoms is of paramount importance when considering possible thrombolytic and interventional therapy.
   c. Definitive care for ischemic stroke is reperfusion therapy, whether it be a systemic thrombolytic given intravenously, local thrombolytic given intra-arterially, or clot retrieval performed via endovascular intervention.

IV. **PHYSICAL EXAM**
   a. Air Care Team members must be able to perform a brief but accurate neurologic examination using the Cincinnati Stroke Scale and/or the National Institutes of Health Stroke Scale (NIHSS) and establish a neurologic baseline should the patient's condition improve or deteriorate.

Cincinnati Stroke Scale

CPSS	**Normal**	**Abnormal**
Facial Droop *The patient shows teeth or smiles.*	Both sides of face move equally.	One side does not move as well as other.
Arm Drift *The patient closes their eyes and extends both arms straight for 10 seconds.*	Both arms move the same, or both arms do not move at all.	One arm either does not move, or one arm drifts down compared to the other.
Speech *The patient repeats, "The sky is blue in Cincinnati".*	The patient says correct words with no slurring of words.	The patient slurs words, says the wrong words, or is unable to speak.

V. **DEPARTMENT GUIDELINE:**

a. Establish care as defined by *GUID3203-G002 – General Patient Care*.
b. Complete a physical neurological exam to include time of onset of symptoms, pupil size and reaction, and Cincinnati Stroke Scale and/or the NIHSS.
   i. If at a sending facility, obtain baseline NIHSS score and monitor for neurological changes.
   i. Examination should include detecting extracranial causes and distinguishing stroke from stroke mimics (i.e. hypoglycemia, seizure, systemic infection, brain tumor, toxic/metabolic disorders, positional vertigo, conversion disorder).
c. Administer oxygen via nasal cannula if O2 saturation is < 92% or if respiratory complaints.
d. Secure peripheral IV access. (Preferred large-bore access in the antecubital fossa for CT scan).
e. Keep HOB at 30 degrees and neck midline if possible.
f. If necessary, secure an advanced airway *per GUID3203-G008 Rapid Sequence Induction for Endotracheal Intubation and GUID3203-PR020 Endotracheal Intubation* and ventilate with a transport ventilator per *GUID3203-PR024 Ventilation with a Mechanical Ventilator*.
   i. Consider short-acting induction and muscle relaxant agents.
   ii. Try to avoid PEEP >5 cm H2O to minimize increases in ICP.
   iii. Ventilate to maintain PaCO2 35-40 mmHg.
g. Perform a glucose check and if glucose is < 60 mg/dL, administer medications per *GUID3203-M021 Hypoglycemia*.
h. If SBP < 90 mmHg, MAP < 65 mmHg, or signs of inadequate perfusion, begin resuscitation with 0.9% NS per *GUID3203-M011 - Hypotension*.
i. Obtain family/contact information for the person who last saw the patient normally for the receiving facility in case additional information is needed regarding the history of events or for a relative who can assist with medical decision making.
j. Hypertensive Blood Pressure Management
   i. For <u>scene</u> *suspected* Ischemic or Hemorrhagic Stroke

1. Allow permissive hypertension up to SBP 220 mmHg until CT scan confirms diagnosis.

ii. For <u>inter-facility</u> *confirmed* stroke (transfers)

1. Confirmed Hemorrhagic Stroke
    a. Goal SBP 120-140 mmHg and MAP > 65 mmHg
        i. **Nicardipine infusion at 5mg/h.** Increase by 2.5mg/h IV every 5-15 minutes to maximum of 15mg/h. titrate to goal above
        ii. **Labetalol 10-20 mg IV** over 2 minutes. May repeat every 10 minutes until goal blood pressure obtained or maximum total dose of 300mg. Avoid if HR < 60 bpm.
    b. Consider pain medication per *GUID3203-G005 Analgesia and Sedation Management* and apply ETCO2 monitoring.
    c. If ICP monitor in place, maintain CPP between 70-100 mmHg.
2. Confirmed Ischemic Stroke
    a. *If tPA is NOT administered:* Permissive hypertension SBP up to 220 mmHg
    b. *If tPA is administered:* Goal SBP 140-180 mmHg / DBP<105
        i. **Nicardipine infusion at 5mg/h.** Increase by 2.5mg/h IV every 5-15 minutes to maximum of 15mg/h. titrate to goals above
        ii. **Labetalol 10-20 mg IV** over 2 minutes. May repeat every 10 minutes until goal blood pressure obtained or maximum total dose of 300mg. Avoid if HR < 60 bpm.
    c. If hypertension persists, contact receiving physician or treat patient per *GUID3203-M010 – Hypertensive Emergency*.
    d. Administer vasopressors for hypotension per *GUID3203-M011 – Hypotension.*

k. If patient is receiving tPA:
- i. Review orders with sending RN and continue infusion en-route
- ii. Monitor for signs/symptoms/complications, specifically intracranial hemorrhage
- iii. *Discontinue* tPA if:
    1. Development of NEW severe headache
    2. Development of significant bleeding (observed)
    3. Development of angioedema
    4. Development of sudden hypotension (suggesting internal hemorrhage)
    5. Blood Pressure > 180/105 despite aggressive intervention
    6. Sudden change in mental status defined as decrease in GCS of ≥ 3 points from baseline eye and verbal OR > 2 points for motor response
- iv. Flush tPA infusion tubing at the end of infusion to administer the complete dose of thrombolytic with 0.9% NS.

l. For hemorrhagic stroke patients taking anticoagulants
- i. Attempt to determine timing of last anticoagulant dose

ii. Obtain copies of appropriate baseline coagulation studies such as: PT, aPTT, INR, TEG, anti-Xa
iii. Continue appropriately ordered reversal agent for patients on anticoagulant medications (anti-inhibitor coagulant complex [FEIBA], idarucizumab [Praxbind], Vitamin K, coagulation factor Xa recombinant [Andexanet alpha]) per transferring or receiving facility orders.
iv. For patients taking Warfarin with an INR > 1.4, consider with sending physician
    1. **Vitamin K 1-2.5 mg PO or 10 mg IV** if no prosthetic heart valve
    2. **FEIBA 1000 units/20 ml IV** over 15 minutes
    3. **Fresh frozen plasma 2 units IV** over 15 minutes
v. For patients taking oral factor Xa Inhibitors such as Apixaban (Eliquis), Edoxaban (Sayvaysa), or Rivaroxaban (Xarelto), consider with sending physician
    1. **FEIBA 2000 units/40 ml IV** over 15 minutes
    2. **Tranexamic acid 1 gram IV** over 10 minutes
vi. If FEIBA was ordered at sending facility:
    1. Wait for the FEIBA medication before transferring the patient. If the patient is at the point of herniating, call the receiving physician to discuss if they would prefer you to transfer without delay.
    2. The IV access used for FEIBA is dedicated to only FEIBA for time of administration.
    3. Ensure the MD/RN at receiving facility is aware of the time that FEIBA was given.
    4. Observe for infusion reactions, thromboembolic events, & allergic reactions.
vii. For patients taking Dabigatran (Pradaxa), consider with sending physician
    1. **Idarucizumab 2.5 grams IV** every 10 minutes x 2 doses

m. For evidence of significantly elevated intracranial pressure and impending cerebral herniation:
   i. If patient has GCS $\leq$ 8 PLUS at least one of the following:
      1. Unilateral fixed and dilated pupil
      2. Unilateral paralysis
      3. Decerebrate posturing
         a. Administer Hypertonic 3% Saline
            i. Adult: **3% NS 500 ml IV** over 10-30 minutes
         b. If interfacilty and sending facility prefers, consider obtaining and administering **Mannitol 1 gram/kg IV** over 5-10 minutes using a filter on IV tubing if adequate blood pressures.
         c. Controlled hyperventilation can be considered to decrease ETCO2 30-35 mmHg
         d. For active seizures, treat per *GUID3203-M015 – Seizures*.
            i. If there are no active seizures and time permits, administer:
               1. Adult: **Levetiracetam 1 gram IV** over 2-5 minutes as a prophylactic anticonvulsant

n. Transport patient to the nearest Comprehensive Stroke Center (as certified by The Joint Commission or DNV GL).

i. Interfacility transport of patients known to require thrombectomy or scene calls with high likelihood of large vessel occlusion may be transported to the nearest *thrombectomy-capable primary stroke center* if there are operational barriers to reaching a Comprehensive Stroke Center.
    1. <u>Central Florida Comprehensive Stroke Centers:</u> Advent Health-Orlando, Advent Health-Tampa, Bayfront Health St. Petersburg, Brandon Regional Hospital, Lakeland Regional Medical Center, Ocala Regional Medical Center, Orlando Regional Medical Center, Osceola Regional Medical Center, UF Health Shands Gainesville, St. Joseph's Hospital Tampa, Tampa General Hospital
    2. <u>Central Florida Thrombectomy Capable Stroke Centers:</u> Halifax Medical Center, Holmes Regional Medical Center, Winter Haven Hospital

ii. If patient meets both Trauma Alert and Stroke Alert criteria, transport to a facility which is a both a Comprehensive Stroke Center and State Approved Trauma Center.

iii. Any patient with suspicion for ruptured aneurysm or subarachnoid hemorrhage (thunderclap headache, altered mental status, non-focal neuro exam) should be transported to the nearest Comprehensive Stroke Center

o. Notify receiving facility as soon as possible with a "Stroke Alert" to activate stroke team.

p. If seizures are present, treat per *GUID3203-M015 - Seizures.*

q. Treat nausea/vomiting per *GUID3203-G004 - Motion Sickness/Nausea.*

r. Avoid D5W or dextrose-based fluids.

VI. **DOCUMENTATION:** EMS Charts

VII. **REFERENCES:**

a. Cook AM, Jones GM, Hawryluk GWJ, et al. Guidelines for the Acute Treatment of Cerebral Edema in Neurocritical Care Patients. *Neurocritical Care.* 2020.

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c. Holbrook A, Schulman S, Witt DM, et.al. Evidence-based management of anticoagulant therapy. Antithrombotic therapy and prevention of thrombosis, 9th Ed: American College of Chest Physicians evidence-based clinical practice guidelines. *Chest.* 2012;141(2)(Suppl):e152S-184S.

d. Meyran, D., Cassan, P., Avau, B., Singletary, E., & Sideman, D. A. (2020). Stroke recognition for first air providers: A systematic review and meta-analysis. *Cureus*. Nov; 12(11): e11386.

e. Warfarin Reversal. ORMC Surgical Critical Care Guideline. Updated 11/29/2017.